Two partial-summation bounds for the divisor function alpha^{omega(n)}

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Editorial note. The counted statement is the author's labelled proposition 'prel:partial summation bounds' (source lines 1678-1685); the proof environment begins on the next line after \end{prop} apart from one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's strong Mertens estimate, whose label is invoked inside the retained proof; retaining it keeps every label used inside the retained text defined inside the package.

Editorial note. Two citations inside the retained proof resolve against the publisher bibliography retained with this source.

Editorial note. No counter adjustment is retained: the publisher numbering is not reproduced and the item prints with the wrapper's own sequential numbering.

Editorial scope. One original result of the pinned publisher article, counted once: the two partial-summation bounds for alpha^{omega(n)} (author label prel:partial summation bounds).

Allowed standard prerequisites (editorial).

- Multiplicative structure of alpha^{omega(n)}. alpha^{omega(n)} is multiplicative, so the Dirichlet series-type sums over n with all prime factors bounded factor as Euler products.. The case y >= sqrt(x) writes the sum as a product over primes p <= y^3 at source line 1691.
- Strong Mertens estimate. sum over prime powers p^k <= x of 1/(k p^k) = log log x + gamma + O(1/(log x)^3).. It is used at source line 1693 to control the Euler product.
- Dyadic summation. A sum over an interval may be split into dyadic blocks (e^{m-1}, e^m] and bounded by the number of blocks times the largest block.. It is the technique of the case y <= sqrt(x) at source lines 1695-1699.

Proposition 0.1 (Strong Mertens's estimate)**.** *For $x \geq 2$, we have*

$$\sum_{p^k \leq x} \frac{1}{kp^k} = \log_2 x + \gamma + O(1/(\log x)^3).$$

Proposition 0.2. *Uniformly for $\alpha \in [0,1]$ and $x, y \geq 2$, we have*

$$\sum_{n \leq x} \alpha^{\omega(n)} \ll x(\log x)^{\alpha-1} \quad \textit{and} \quad \sum_{x/y < n \leq xy} \frac{\alpha^{\omega(n)}}{n} \ll (\log y)(\log(xy))^{\alpha-1}.$$

Proof. The first bound follows readily with [13, Th. 14.2]. Let us now prove the second one.

When $y \geq \sqrt{x}$, we have

$$\sum_{x/y < n \leq xy} \frac{\alpha^{\omega(n)}}{n} \leq \sum_{p|n \ \Rightarrow p \leq y^3} \frac{\alpha^{\omega(n)}}{n} = \prod_{p \leq y^3} \Big(1 + \frac{\alpha}{p-1}\Big) \ll (\log y)^{\alpha}$$

by the inequality $1 + t \leq e^t$ and Mertens's theorem (Proposition 0.1). This completes the proof in this case. Finally, assume that $y \leq \sqrt{x}$. We then have

$$\sum_{x/y < n \leq xy} \frac{\alpha^{\omega(n)}}{n} \leq \sum_{\substack{m \in \mathbb{Z} \\ e^m \in [x/y, exy]}} \sum_{n \in (e^{m-1}, e^m]} \frac{\alpha^{\omega(n)}}{n} \leq \sum_{\substack{m \in \mathbb{Z} \\ e^m \in [x/y, exy]}} e^{1-m} \sum_{n \leq e^m} \alpha^{\omega(n)}.$$

For every m as above, the innermost sum is $\ll e^m m^{\alpha-1} \asymp e^m (\log(xy))^{\alpha-1}$ by the first part of the lemma and our assumption that $y \leq \sqrt{x}$. Since there are $\leq 1 + \log y \ll \log y$ choices for m, the needed estimate follows in this last case too. □

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